

Rudenko Varvara

Head of ML department/ Researcher

✉ Rudenko.VD@phystech.edu

🖱️ scopus.com

🖱️ scholar.google.com

🔄 github.com/Rudenshtok

Education

2026 – 2029 (expected date)

MIPT PhD

Department of Control and Applied Mathematics

2024 – 2026

MIPT Master's degree

Department of Control and Applied Mathematics

GPA: 5.0/5.0

Thesis: "Asynchronous Policy Gradient Aggregation for Efficient Distributed Reinforcement Learning"
Advisor: Radoslav Neychev, PhD

2019 – 2023

MIPT Bachelor's degree

Department of Control and Applied Mathematics

GPA: 4.5/5.0

Thesis: "Regret Minimization for Discounted MDPs by Variance-Reduced Q-learning"
Advisor: Prof. Alexander Gasnikov

Technical Skills

</> **Languages:** Python (research), C++ (basic), SQL (basic)
</> **Python:** Research scripting, PyTorch prototyping, data analysis
</> **ML/RL:** NumPy, SciPy, Gymnasium, Stable Baselines3 (basic)
</> **Tools:** Git, LaTeX, Linux, Docker (basic), Postman, Jupyter Notebook, REST APIs
</> **Math:** MDPs, stochastic optimization, convex analysis, MCMC

English

C1

Research Profile

My research focuses on **sample-efficient reinforcement learning for Markov Decision Processes** and **stochastic optimization for multi-agent systems**. I am particularly interested in developing theoretically-grounded algorithms with convergence guarantees for real-world applications.

Publications

Asynchronous Policy Gradient Aggregation for Efficient Distributed Reinforcement Learning

A Tyurin, A Spiridonov, **V Rudenko**

🖱️ ICLR.cc

ICLR 2026

Algorithm for Constrained Markov Decision Process with Linear Convergence

E Gladin, M Lavrik-Karmazin, K Zainullina, **V Rudenko**, A Gasnikov, Martin Takac

🖱️ Arxiv.org

AISTATS 2023

Markov Decision processes and convex optimization

VD Rudenko, NE Yudin, AA Vasin

🖱️ CRM.org

CRM 2023

Parallel Stochastic Cubic Newton with Dual Momentum in Multi-agent Optimization setup

First author, under review

In preparation

Conferences & Talks

🖱️ 65th All-Russian Scientific Conference of MIPT 2023 "Markov Decision processes and convex optimization"

🖱️ Fall into ML 2023 "ALGORITHM FOR CONSTRAINED MARKOV DECISION PROCESS WITH LINEAR CONVERGENCE"

Grants

- ▶ Recognized by international quantitative finance industry (offer from XTX Markets, 2026)
- ▶ Top 100 MIPT. Winner of the "Research Breakthrough" category.
- ▶ High Professional Potential Group (High School Economics University Personnel Reserve) Category "New researchers"
- ▶ Laureate of the scholarship for achievements in the field of numerical optimization methods MIPT
- ▶ Optimization and control methods in the complex robotic systems with many degrees of freedom in a dynamic environment Sber Allab

Additional Education

2024

Artificial intelligence in climate sciences

Sirius University of Science and Technology

2022

Modern methods of optimization

Sirius University of Science and Technology

2021

Fundamentals of Reinforcement Learning

The Alberta Institute & Alberta Machine Intelligence Institute

2020, 2021

Sampling, management and optimization & Stochastic algorithms and machine learning

Sirius University of Science and Technology

References

Prof. Martin Takac

Email: martin.takac@mbzuai.ac.ae

Radoslav Neychev, PhD

Email: neychev@phystech.edu

Charitable activity

2022 – 2024

Teaching mathematics at a Charity Fund "Gift of life" for cancer-stricken children

Work experience

Head and Leading Specialist in Machine Learning

2025 – today

Joint Stock Company "Channel One"

1tv.ru

Development of a complete AI agent architecture and orchestration tools for the First Channel website. Leading a team of developers and editors to integrate new neural network technologies, reducing manual workflows by 75%.

Lecturer

2023 – today

Reinforcement Learning course MIPT University/ Math statistics course HSE University/ MIPT Olympiad school

Developing materials and teaching the course as a lecturer

Research Associate

2024 – 2025

Artificial Intelligence Research Institute (AIRI)

Airi.net

Working on writing articles and obtaining new results in the field of stochastic optimization for MDP

ICLR 2026 Asynchronous Policy Gradient Aggregation for Efficient Distributed Reinforcement Learning

Research Associate

2021 – 2025

Laboratory of mathematical methods of optimization

LabMMO.ru

Robust Submodular Multi-Agent Reinforcement Learning Algorithms – in preparation

2025 Multi-agent Reinforcement Learning system in double auction setting – Creation of a MARL user environment in which several agents trade with each other in a continuous double auction

2024 Determination of garbage pollution from satellite photos – Work is underway to refine the method for detecting debris in satellite photographs in a small dataset

CRM 2023 Markov Decision processes and convex optimization

Research Associate

2020 – 2024

International Laboratory of Stochastic Algorithms and Multidimensional Data Analysis

HDILab.ru

AISTATS 2023 Algorithm for Constrained Markov Decision Process with Linear Convergence

2023 Regret Minimization for Discounted MDPs by Variance-Reduced Q-learning – New adaptation of classical Q-learning algorithm and introduce Q-learning with Hoeffding bonuses and Bernstein Bonuses

2022 Langevin Monte Carlo: random coordinate descent and variance reduction – The idea was to improve the ULA algorithm using a randomized coordinate vector

2021 UVIP: Model-Free Approach to Evaluate Reinforcement Learning Algorithms – Work with KBRL, KBSF methods, testing algorithm for two Gaussians

2020 Importance Sampling and control variates – Improving estimates by adding control variates with zero variance